

I am confused about useParams and useSearchParams in react

Here's a clear, simple explanation of **useParams** vs **useSearchParams** in React Router (v6+).

✅ useParams

What it reads: Route **path parameters** — the dynamic segments *inside the URL path*.

Where: Defined in the route pattern.

Example URL:

```
bash
```

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```
/users/123
```

Route definition:

```
jsx
```

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```
<Route path="/users/:id" element={<User />} />
```

In the component:

```
jsx
```

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```
import { useParams } from "react-router-dom";
```

```
const User = () => {  
  const { id } = useParams();  
  console.log(id); // "123"  
}
```

✅ When to use useParams

- You need **required, structured** data from the URL
- Part of navigation (e.g., `/posts/:postId`)
- Works only with values defined in the route path

useSearchParams

What it reads: URL query string parameters — after the `?`.

Example URL:

bash

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```
/users?id=123&sort=asc
```

In the component:

jsx

Copy code

```
import { useSearchParams } from "react-router-dom";

const Users = () => {
  const [searchParams] = useSearchParams();
  const id = searchParams.get("id");
  const sort = searchParams.get("sort");

  console.log(id); // "123"
  console.log(sort); // "asc"
}
```

When to use useSearchParams

- You need **optional** filtering/sorting values
- Useful for things like:
 - pagination `?page=2`
 - search filters `?q=shoes&color=red`
 - toggles or UI state saved in the URL

Bonus: You can update search params:

jsx

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```
const [searchParams, setSearchParams] = useSearchParams();

setSearchParams({ page: 2, sort: "desc" });
```

Key Difference Summary

Feature	<code>useParams</code>	<code>useSearchParams</code>
Comes from	Route path	URL query string
Looks like	<code>/users/:id</code>	<code>/users?id=123</code>
Returns	Object of dynamic segments	<code>URLSearchParams</code> instance
Best for	Identifying resources	Filtering, searching, sorting
Optional?	Usually required	Usually optional

Quick way to remember

- `useParams` → structural, part of the path
- `useSearchParams` → optional, part of the query

If you want, I can show examples with nested routes, updating search params, or converting search params to state!

